MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

01

10 answers · Rationale-rich · Teach from doc

Q1

11/4

Accept also: `2.75`

The correct answer is $\frac{11}{4}$. It's given that the function f is defined by $fx = 5\frac{1}{4} - x^{2} + \frac{11}{4}$. Substituting $\frac{1}{4}$ for x in this equation yields $f\frac{1}{4} = 5\frac{1}{4} - \frac{1}{4}^{2} + \frac{11}{4}$, which is equivalent $f\frac{1}{4} = 50^{2} + \frac{11}{4}$, or $f\frac{1}{4} = \frac{11}{4}$. Therefore, the value of $f\frac{1}{4}$ is $\frac{11}{4}$. Note that 11/4 or 2.75 are examples of ways to enter a correct answer.

Q2

A

A. $x + 15^{2} = 0$

Choice A is correct. Taking the square root of both sides of $(x + 15)^{2} = 0$ yields $x + 15 = 0$. Subtracting 15 from both sides of this equation yields $x = -15$. Therefore, this equation has exactly one distinct real solution.

Choice B is incorrect. This equation has no distinct real solutions, not exactly one distinct real solution.

Choice C is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Choice D is incorrect. This equation has exactly two distinct real solutions, not exactly one distinct real solution.

Q3

B

B. $b^2 - 2bc + c^2$

Choice B is correct. It's given that $x^2 = a + b$ and $y^2 = a + c$. Using the distributive property, the expression $(x^2 - y^2)^2$ can be rewritten as $(x^2)^2 - 2x^2y^2 + (y^2)^2$. Substituting $a + b$ and $a + c$ for x^2 and y^2, respectively, in this expression yields $(a+b)^2 - 2((a+b)(a+c)) + (a+c)^2$. Expanding this expression yields $(a^2 + 2ab + b^2) - (2a^2 + 2bc + 2ac + 2ab) + (a^2 + 2ac + c^2)$. Combining like terms, this expression can be rewritten as $b^2 - 2bc + c^2$.

Choices A, C, and D are incorrect and may result from an error in using the distributive property, substituting, or combining like terms.

Q4

B

B. This toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

Choice B is correct. The vertex of the graph of a quadratic equation is where it reaches its minimum or maximum value. When a quadratic equation is written in the form $y = a(x - h)^2 + k$, the vertex of the parabola represented by the equation is $(x, y) = (h, k)$. In the given equation $y = -16(x - 5.6)^2 + 502$, the value of h is 5.6 and the value of k is 502. It follows that the vertex of the graph of this equation in the xy-plane is $(x, y) = (5.6, 502)$. Additionally, since $a = -16$ in the given equation, the graph of the quadratic equation opens down, and the vertex represents a maximum. It's given that the value of y represents the estimated height, in feet, of the toy rocket x seconds after it is launched into the air. Therefore, this toy rocket reaches an estimated maximum height of 502 feet 5.6 seconds after it is launched into the air.

Choice A is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice C is incorrect. The 16 in the equation is an indicator of how narrow the graph of the equation is rather than where it reaches its maximum.

Choice D is incorrect. This is an interpretation of the vertex of the graph of the equation $y = -16(x - 502)^2 + 5.6$, not of the equation $y = -16(x - 5.6)^2 + 502$.

Q5

A

A. $4j + 9 = \frac{k}{p}$

Choice A is correct. To express $4j + 9$ in terms of p and k, the given equation must be solved for $4j + 9$. Since it's given that j is a positive number, $4j + 9$ is not equal to zero. Therefore, multiplying both sides of the given equation by $4j + 9$ yields the equivalent equation $p4j + 9 = k$. Since it's given that p is a positive number, p is not equal to zero. Therefore, dividing each side of the equation $p4j + 9 = k$ by p yields the equivalent equation $4j + 9 = \frac{k}{p}$.

Choice B is incorrect. This equation is equivalent to $p = \frac{4j + 9}{k}$.

Choice C is incorrect. This equation is equivalent to $p = k - 4j - 9$.

Choice D is incorrect. This equation is equivalent to $p = k4j + 9$.

Q6

C

C. $\frac{1}{9}$

Choice C is correct. Multiplying the given functions f and g yields $fx \cdot gx = x^2 + bx9x^2 - 27x$. Applying the distributive property to the right-hand side of this equation yields $fx \cdot gx = x^2 9x^2 - 27x + bx9x^2 - 27x$. Applying the distributive property once again to the right-hand side of this equation yields $fx \cdot gx = x^2 9x^2 - x^2 27x + bx9x^2 - bx27x$, which is equivalent to $fx \cdot gx = 9x^4 - 27x^3 + 9bx^3 - 27bx^2$. Factoring out x^3 from the second and third terms yields $fx \cdot gx = 9x^4 + -27 + 9bx^3 - 27bx^2$. Since the left-hand sides of $fx \cdot gx = 9x^4 + -27 + 9bx^3 - 27bx^2$ and $fx \cdot gx = 9x^4 - 26x^3 - 3x^2$ are equal, it follows that $-27 + 9bx^3 = -26x^3$, or $-27 + 9b = -26$, and $-27bx^2 = -3x^2$, or $-27b = -3$. Adding 27 to each side of $-27 + 9b = -26$ yields $9b = 1$. Dividing each side of this equation by 9 yields $b = \frac{1}{9}$. Similarly, dividing each side of $-27b = -3$ by -27 yields $b = \frac{1}{9}$. Therefore, the value of b is $\frac{1}{9}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7

A

A. The estimated initial beetle population for this species and area in 2012

Choice A is correct. For an exponential function in the form $fx = ab^x$, where a and b are positive constants and $b < 1$, the initial value of $f(x)$, or the value of $f(x)$ when $x = 0$, is a and the value of $f(x)$ decreases by $100(1 - b)\%$ each time x increases by 1. Therefore, the initial value of the function $fx = 4{,}0000.75^x$, or the value of $f(x)$ when $x = 0$, is 4,000. Therefore, the best interpretation of 4,000 in this context is the estimated initial beetle population for this species and area in 2012.

Choice B is incorrect. The estimated beetle population for this species and area 7 years after 2012 is $4{,}000(0.75)^7$, or approximately 534, not 4,000.

Choice C is incorrect. The estimated percent decrease in the beetle population for this species and area each year after 2012 is $100(1 - 0.75)$, or 25, not 4,000.

Choice D is incorrect. The estimated percent decrease in the beetle population for this species and area every 7 years after 2012 is $100\left(1 - 0.75^7\right)$, or approximately 87, not 4,000.

Q8

either 0 or 3

The correct answer is either 0 or 3. For an ordered pair to satisfy a system of equations, both the x- and y-values of the ordered pair must satisfy each equation in the system. Both expressions on the right-hand side of the given equations are equal to y, therefore it follows that both expressions on the right-hand side of the equations are equal to each other: $x^2 - 4x + 4 = 4 - x$. This equation can be rewritten as $x^2 - 3x = 0$, and then through factoring, the equation becomes $x(x - 3) = 0$. Because the product of the two factors is equal to 0, it can be concluded that either $x = 0$ or $x - 3 = 0$, or rather, $x = 0$ or $x = 3$. Note that 0 and 3 are examples of ways to enter a correct answer.

Q9 B

B. $2a+5b$

Choice B is correct. The first and last terms of the polynomial are both squares such that $4a^2=(2a)^2$ and $25b^2=(5b)^2$. The second term is twice the product of the square root of the first and last terms: $20ab=2(2a)(5b)$. Therefore, the polynomial is the square of a binomial such that $4a^2+20ab+25b^2=(2a+5b)^2$, and $(2a+5b)$ is a factor.

Choice A is incorrect and may be the result of incorrectly factoring the polynomial. Choice C is incorrect and may be the result of dividing the second and third terms of the polynomial by their greatest common factor. Choice D is incorrect and may be the result of not factoring the coefficients.

Q10 B

B. The diver hits the water at 1.6 seconds.

Choice B is correct. It's given that the graph shows the height above the water y, in meters, of a diver x seconds after diving from a platform. The x-intercept of a graph is the point at which the graph intersects the x-axis, or when the value of y is 0. The graph shown intersects the x-axis between $x = 1$ and $x = 2$. In other words, the diver is 0 meters above the water, or hits the water, between 1 and 2 seconds after diving from the platform. Of the given choices, only choice B includes an interpretation where the diver hits the water between 1 and 2 seconds. Therefore, the best interpretation of the x-intercept of the graph is the diver hits the water at 1.6 seconds.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the best interpretation of the maximum value, not the x-intercept, of the graph.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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